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# Conflict-aware semi-supervised mutual learning for medical image segmentation

Wenlong Hang [a], Beijing Wang [a], Shuang Liang [b], Qingfeng Zhang [c], Qiang Wu [c,*], Yukun Jin [c], Qiong Wang [d,*], Jing Qin [e]

[a] College of Computer and Information Engineering/College of Artificial Intelligence, Nanjing Tech University, Nanjing, 211816, China
[b] School of Internet of Things, Nanjing University of Posts and Telecommunications, Nanjing, 210023, China
[c] Department of Otorhinolaryngology Head and Neck Surgery, Shenzhen University General Hospital, Shenzhen, 518055, China
[d] Guangdong Provincial Key Laboratory of Computer Vision and Virtual Reality Technology, Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences, Shenzhen, 518055, China
[e] School of Nursing, Hong Kong Polytechnic University, Hong Kong, China

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ABSTRACT

Semi-supervised learning (SSL) has shown promising performance in medical image segmentation by effectively utilizing extensive unlabeled images. However, inaccurate predictions of unlabeled images can significantly impair the segmentation performance of SSL models. Furthermore, most current SSL methods lack mechanisms to handle cognitive bias, causing the model easily overfit on inaccurate predictions and making self-correction challenging. In this work, we propose a conflict-aware semi-supervised mutual learning framework (CSSML), which integrates two different subnetworks and selectively utilizes conflicting pseudo-labels for mutual supervision to address these challenges. Specifically, we introduce two subnetworks with different architecture incorporating a conflict-aware distinct feature learning (CDFL) regularization to avoid the homogenization of subnetworks while promoting diversified predictions. To handle potential inaccurate predictions, we introduce a geometry-aware mutual pseudo supervision (GMPS) regularization to determine the reliability of conflicting pseudo-labels of unlabeled images, and selectively leverage the more reliable pseudo-labels in the two subnetworks to supervise the other one. The synergistic learning between CDFL and GMPS regularizations during the training process facilitates each subnetwork to selectively incorporates reliable knowledge from the other subnetwork, thereby helping the model overcome cognitive bias. Extensive experiments on three public medical image datasets demonstrate that the proposed CSSML achieves an average of 80.65% DSC, 87.83% Precision, and 14.48mm 95HD using only 20% labeled data, highlight-ing its superior performance. The code is available at: https://github.com/Mwnic-AI/CSSML.

## 1. Introduction

Automated image segmentation plays a crucial role in medical image analysis, serving as a fundamental component for computer-aided diagnosis, treatment planning, and interventional procedures. Accurate medical image segmentation enables the quantitative analysis of tissue morphological attributes, thus enhancing its practicality in supporting medical decision-making processes Zhao et al. (2023), Zhou et al. (2023). Recent advancements in convolutional neural networks (CNNs) have significantly improved automated medical image segmentation methods, achieving and, in certain instances, surpassing human-level performance in various complex clinical tasks Bortsova et al. (2019), Greenwald et al. (2022). The efficacy of CNN models is highly dependent on the availability of substantial quantities of labeled images. Nevertheless, the acquisition of extensive pixel-wise annotations for medical images is often labor-intensive, time-consuming, and requires specialized medical expertise. In clinical practice, the collection of large quantities of unlabeled medical images is relatively convenient. In light of this, the semi-supervised learning (SSL) paradigm, which enables models to leverage abundant unlabeled data to mitigate the burden of pixel-wise annotation, has progressively emerged as the mainstream paradigm for medical image segmentation tasks Adiga et al. (2024).

The success of the SSL paradigm is attributed to its ability to concurrently exploit knowledge from both labeled and unlabeled images. In general, current SSL methods mainly adopt two strategies to utilize unlabeled data, *i.e.*, consistency regularization (CR) Luo et al. (2021), Yu et al. (2019) and pseudo-labeling (PL) Rizve et al. (2021), Su et al. (2024). CR-based SSL methods aim to encourage models to produce

* Corresponding authors.
*E-mail addresses:* sdzywq@szu.edu.cn (Q. Wu), wangqiong@siat.ac.cn (Q. Wang).

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invariant predictions for input images subjected to minor perturbations at the data, feature, and network levels Liu et al. (2022). PL-based SSL methods predominantly follow the self-training paradigm, which involves assigning pseudo-labels to unlabeled data and subsequently integrating the pseudo-labeled data with the labeled data to retrain the model. Furthermore, some SSL methods Chen et al. (2021b), Ouali et al. (2020) have been proposed to combine the CR and PL strategies to improve segmentation performance.

Despite the considerable success of current SSL methods, one potential weakness is that inaccurate predictions of unlabeled data can significantly degrade their performance. For CR-based SSL methods, perturbations applied to unlabeled data with inaccurate predictions can deteriorate the consistency learning performance Liu et al. (2022). Additionally, inaccurate pseudo-labels inevitably introduce unreliable pseudo-labeled data into the training set, thereby misleading the training process for PL-based SSL methods Su et al. (2024). Most SSL methods utilize two subnetworks with identical architectures to make predictions for unlabeled data. The homogenization of subnetworks constrains prediction diversity for unlabeled data, making it difficult for them to rectify each other's errors. This causes the model to easily overfit to inaccurate predictions and makes self-correction difficult Li et al. (2023). Nonetheless, most SSL methods lack mechanisms to address the aforementioned issues. For instance, mean teacher (MT)-based Tarvainen and Valpola (2017) SSL methods employ two subnetworks (referred to as the teacher and student) with identical architectures, encouraging them to produce invariant predictions for the same inputs under perturbations. Similarly, the deep co-training (DCT)-based SSL method Qiao et al. (2018) encourages consistent predictions for the same input across different views. Besides, cross pseudo supervision (CPS)-based SSL methods Chen et al. (2021b) employ two subnetworks with identical architecture but different initializations to generate pseudo-labels for cross supervision. Although effective, the homogenization of subnetworks hinders the efficiency of knowledge exchange between them, thereby constraining the potential benefits of multi-subnetwork architectures. Consequently, these methods remain susceptible to cognitive bias and potential model collapse caused by unreliable pseudo-labels Yang et al. (2022).

To address the aforementioned issues, some SSL methods employ subnetworks with distinct architectures to prevent them from becoming overly coupled. In Li et al. (2023), a diverse co-training framework was developed to leverage multiple subnetworks with distinct architectures and input domains to generate diversified pseudo-labels for cross supervision. Luo et al. (2022) utilized a CNN and a transformer Vaswani et al. (2017) as the subnetworks to facilitate cross teaching between them. Furthermore, Wang et al. (2023a) introduced a mutual correction framework that integrated two distinct subnetworks and employed a dynamic competitive pseudo-label generation scheme to select the more reliable outputs from the two subnetworks for supervising the training of the other. However, medical images contain a variety of lesions that exhibit uneven intensities, adjacent boundaries, and irregular shapes. Despite some advancements in recent years, these SSL methods still confront several significant challenges in medical image processing. Firstly, different subnetworks may still generate identical incorrect predictions for regions prone to missegmentation, such as edge regions. Secondly, these methods generally ignore the adverse effects of unreliable pseudo-labels. For instance, Li et al. (2023) employed a confidence threshold to filter out unreliable pseudo-labels with low confidence for cross supervision. Nonetheless, determining an appropriate predefined threshold is challenging, as an improper threshold can potentially degrade model performance. In Wang et al. (2023a), the Dice loss was utilized to quantify the segmentation performance of subnetworks and to select the superior outputs as supervision. However, this approach fails to account for potential inaccuracies in pixel-level pseudo-labels during the training procedure.

To alleviate the above two issues, we propose a novel conflict-aware semi-supervised mutual learning framework, termed CSSML, to facilitate semi-supervised medical image segmentation. To fully exploit the potential of the multi-subnetwork architecture, CSSML employs two subnetworks with distinct structures. To minimize the risk of the subnetworks making identical incorrect predictions in regions prone to missegmentation, we introduce a conflict-aware distinct feature learning (CDFL) regularization. CDFL regularization aims to reduce the coupling of features extracted by the subnetworks to further promote diversified predictions. Most importantly, CDFL enables the automatic learning of diversified feature representations without relying on explicit perturbations. Additionally, for the pseudo-labels generated by the two subnetworks, we categorize them based on whether they conflict. Identical pseudo-labels indicate that the corresponding pixels are more likely to be accurately predicted. Therefore, we simply select the pseudo-label with the higher probability value from the two subnetworks to supervise its counterpart. For the conflicting pseudo-labels, we introduce a geometry-aware mutual pseudo supervision (GMPS) regularization. GMPS regularization can adaptively identify and utilize more reliable pseudo-labels from two subnetworks to supervise their counterparts, thereby preventing model collapse and enhancing the robustness of mutual learning. The synergistic learning between CDFL and GMPS regularizations during the training process facilitates each subnetwork to selectively incorporate the reliable knowledge imparted by the other subnetwork, thereby helping the model get rid of cognitive bias. In summary, the main contributions of this work are as follows:

- We propose a novel SSL framework to facilitate semi-supervised medical image segmentation by concurrently exploiting conflict-aware subnetworks and addressing conflicting pseudo-labels.
- Our framework incorporates two novel modules: CDFL regularization to promote diversified predictions, and GMPS regularization to selectively leverage the more reliable pseudo-labels from conflicting predictions to supervise the other subnetwork.
- The synergistic learning between the CDFL and GMPS regularizations helps the model overcome cognitive bias. Extensive experimental results on three medical image datasets demonstrate its superiority.

## 2. Related works

### 2.1. Semi-supervised learning

The semi-supervised learning (SSL) paradigm has garnered significant attention as an efficient approach for leveraging limited labeled data alongside abundant unlabeled data. Most current semi-supervised learning (SSL) methods employ either consistency regularization (CR), pseudo-labeling (PL), or a combination of both to effectively utilize unlabeled data. CR-based SSL methods Hang et al. (2020), Liu et al. (2022), Luo et al. (2021, 2022, 2018), Miyato et al. (2018), Ouali et al. (2020), Tarvainen and Valpola (2017) are designed to encourage consistent predictions from subnetworks trained on the same inputs under different types of perturbations. As a pioneering deep SSL framework, MT Tarvainen and Valpola (2017) enforces invariant predictions of student and teacher networks to different input perturbations on the unlabeled data. Building upon the MT framework, numerous advanced SSL methods have been developed by incorporating diverse perturbation techniques to enhance the benefits derived from consistency learning. For instance, the cross-consistency training method Ouali et al. (2020) employs multiple categories of feature perturbations and encourages the concordance between the predictions of perturbed features and those of the non-perturbed features. Interpolation Consistency Training (ICT) Verma et al. (2022) encourages the prediction at an interpolation of unlabeled points to be consistent with the interpolation of the predictions at those points. In Liu et al. (2022), three types of perturbations, *i.e.*, input image perturbation, feature perturbation, and network perturbation, were jointly utilized to enhance model training and further improve the generalization of consistency learning. For PL-based SSL methods Kalluri et al. (2019), Lee et al. (2013), Rizve et al. (2021),

Sohn et al. (2020), Su et al. (2024), Wu et al. (2022), Zeng et al. (2023), Zhang et al. (2021), they primarily follow the self-training paradigm, which is based on the cluster assumption. This assumption suggests that clusters corresponding to each class should exhibit compactness, thereby resulting in low entropy Wu et al. (2022). Most current PL-based SSL methods typically employ a predefined confidence threshold to eliminate low-quality pseudo-labels Li et al. (2023). For example, a confidence threshold Sohn et al. (2020) of 0.95 is utilized to discard unreliable pseudo-labels. However, determining an appropriate threshold that can effectively filter out all unreliable pseudo-labels remains a significant challenge. Some other methods employ a curriculum pseudo-labeling strategy Zhang et al. (2021) or uncertainty-aware pseudo-label selection strategy Rizve et al. (2021) to dynamically filter out unreliable pseudo-labels. Additionally, some recent SSL methods tend to combine CR and PL strategies through cross supervision between subnetworks, which have achieved substantial performance gains in many practical applications Berthelot et al. (2019), Chen et al. (2021b), Sohn et al. (2020).

### 2.2. Cognitive bias in semi-supervised learning

Although the aforementioned SSL methods have achieved promising results, they neglect the impact of cognitive bias within the model. Cognitive bias refers to the phenomenon in which a model persistently makes incorrect predictions due to overfitting to erroneous supervision signals Wang et al. (2023a). Some recent works aim to utilize unified training paradigms Zeng et al. (2025c,d) and self-supervised masking strategies Zeng et al. (2025a) to mitigate this issue. Although effective, the utilization of identical subnetwork architectures and consistency-based pseudo-label generation strategies makes these SSL methods susceptible to cognitive bias, thereby hindering their ability to self-correct. To address these issues, some SSL methods Li et al. (2023), Luo et al. (2022), Wang et al. (2023a) have tended to employ multiple subnetworks with distinct architectures. In Li et al. (2023), two-way and three-way CPS frameworks were developed using a CNN and a transformer as subnetworks. Despite the state-of-the-art performance of these methods, they often ignore the adverse effects caused by unreliable pseudo-labels. Furthermore, Wang et al. (2023a) proposed a mutual correction framework (MCF) that integrates two distinct subnetworks and employs a dynamic competitive pseudo-label generation scheme to select the more reliable outputs from the two subnetworks to supervise the training of the other one. However, MCF conducts cross supervision at the image level, thus overlooking the reliability of pseudo-labels at the pixel level. Building upon MCF, several SSL methods have been developed, including pseudo-label guided selective mutual learning (PLSML) Hang et al. (2025), the learnable prompting SAM-induced knowledge distillation framework (Know-SAM) Huang et al. (2025a), and the uncertainty-aware cross-training framework (UC-Seg) Huang et al. (2025b). Although these approaches have demonstrated effectiveness, they overlook the inconsistent pseudo-labels generated by the two subnetworks, thereby remaining susceptible to cognitive bias. In light of this, we devise GMPS regularization to identify the inconsistent pixel-level predictions and adaptively utilize more reliable ones from both subnetworks to supervise its counterpart, thereby preventing model collapse and enhancing the robustness of model learning.

### 2.3. Semi-supervised medical image segmentation

By leveraging a limited amount of annotated data along-side a substantial quantity of unannotated data, SSL methods have demonstrated promising performance in semi-supervised medical image segmentation across a variety of clinical applications. Along the line of CR strategy, many SSL methods Chen et al. (2021a), Hang et al. (2020), Luo et al. (2021), Xie et al. (2021) have been proposed for medical image segmentation. For example, the uncertainty-aware self-ensembling model (UA-MT) Yu et al. (2019) enforces prediction consistency for the same input under different perturbations. Furthermore, an uncertainty-aware scheme was introduced to enable the student model to learn from reliable targets. In Hang et al. (2020), the local and global structural consistency regularizations were developed to jointly leverage data-level and data-structure-level information within predictions, thereby enhancing the benefits derived from consistency learning. Besides, a dual-task deep network Luo et al. (2021) was introduced to generate predictions for both the pixel-wise segmentation map and the geometry-aware level set representation of the target. Subsequently, a dual-task consistency regularization was employed to ensure coherence between the segmentation maps derived from the level set and those directly predicted. To address the distributional differences across different data domains, Chen et al. (2021a) proposed an adaptive hierarchical dual consistency approach for semi-supervised segmentation on cross-domain datasets. In Zeng et al. (2024), a novel SSL method was proposed to enlarge the discrepancies by training differential decoders, and then learn from the differential medical image features iteratively. In addition to the aforementioned SSL methods, many other learning strategies have been developed to integrate with CR to enhance segmentation performance, such as different perturbation techniques Li et al. (2020a), Shi et al. (2021), Xie et al. (2021) and co-training mechanism Li et al. (2021). PL-based semi-supervised medical image segmentation methods Chaitanya et al. (2023), Li et al. (2020b), Seibold et al. (2022), Yao et al. (2022) usually select a set of predictions to generate pseudo-labels to fine-tune the model. For example, Li et al. (2020b) employed self-generated pixel-wise pseudo-labels as ground truth to expand the training set. A novel confidence-aware cross pseudo supervision method Yao et al. (2022) based on Kullback-Leibler divergence has been developed for semi-supervised domain-generalized medical image segmentation. Additionally, Chaitanya et al. (2023) employed local contrastive loss to acquire effective pixel-level features for segmentation, leveraging semantic label information derived from the pseudo-labels of unlabeled images. Based on the assumption that visually similar regions between labeled and unlabeled images are likely to contain identical semantics, Seibold et al. (2022) employed labeled images as references to generate pseudo-labels for unlabeled images. Other semi-supervised medical image segmentation methods Wu et al. (2021), Yu et al. (2019) tend to integrate both the CR and PL paradigms. For example, Wu et al. (2021) developed a cyclic pseudo-labeling scheme to enforce mutual consistency in challenging regions of medical images, thereby contributing to the minimization of uncertainty in predictions.

Inspired by existing works, we utilize two subnetworks with distinct architectures, incorporating the CDFL regularization to mitigate the issue of cognitive bias. To reduce the impact of inaccurate pseudo-labels, we further devise the GMPS regularization to selectively leverage the more reliable pseudo-labels from conflicting predictions to supervise the other subnetwork.

## 3. Conflict-aware semi-supervised mutual learning framework

In this section, we elucidate the proposed SSL method for medical image segmentation. We first provide a brief introduction to our CSSML framework. Following this, we thoroughly describe the techniques for enhancing prediction diversity and leveraging conflicting predictions for mutual pseudo-supervision, respectively.

### 3.1. Architecture overview

To describe the proposed CSSML more succinctly, we first introduce the basic formulations of SSL. We have a training dataset $D$ that contains $N$ fully pixel-wise annotated medical images $D^l = \left\{\left(\mathbf{x}_i^l, \mathbf{y}_i^l\right)\right\}_{i=1}^{N}$ and $M$ unannotated medical images $D^u = \left\{\mathbf{x}_i^u\right\}_{i=N+1}^{N+M}$. In clinical practice, we have $N \ll M$. $\mathbf{x}_i^l \in \mathcal{X} \subset \mathbb{R}^{H\times W\times C}$ denotes the $i$-th training image in $D^l$, where $H$, $W$, and $C$ denote the height, weight, and color channels of the medical image. $\mathbf{y}_i^l \in \mathcal{Y} \subset \{0,1\}^{H\times W\times Y}$ denotes the corresponding segmentation map of $\mathbf{x}_i^l$, where $Y$ indicates the number of classes.

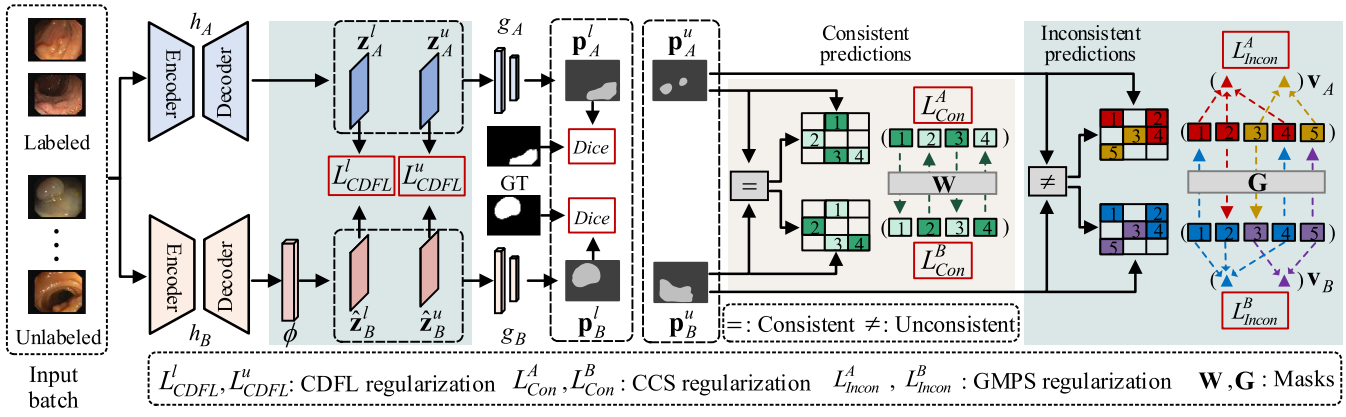



**Fig. 1.** The framework of CSSML. A mini-batch input containing labeled and unlabeled images are fed into subnetworks A and B, respectively. A projection function $\phi$ maps the features extracted by $h_B$ to $\hat{\mathbf{z}}^l_B$ and $\hat{\mathbf{z}}^u_B$, that are used for CDFL regularization with features $\mathbf{z}^l_A$ and $\mathbf{z}^u_A$. The predictions of labeled images $\mathbf{p}^l_A$ and $\mathbf{p}^l_B$ are used to calculate dice loss. The consistent predictions within $\mathbf{p}^u_A$ and $\mathbf{p}^u_B$ are used to calculate CCS regularization, while the inconsistent predictions are used to calculate GMPS regularization. Boxes of different colors correspond to predictions for different classes, with deeper shades indicating higher prediction reliability.

We define the two networks employed in CSSML as subnetwork A ($f_A(\cdot)$) and subnetwork B ($f_B(\cdot)$). We denote $\theta_A$ and $\theta_B$ as the model parameters of subnetworks A and B, respectively. The subnetwork $f_A$ can be decomposed into a feature extractor $h_A(\cdot): \mathcal{X} \to \mathcal{Z}$ and a classifier $g_A(\cdot): \mathcal{Z} \to \mathcal{Y}$, parameterized by $\theta^h_A$ and $\theta^g_A$, respectively. Here, $\mathcal{Z} \subset \mathbb{R}^Z$ represents the feature space of dimension $Z$. Hence, we have $f_A = g_A \circ h_A$ and $\theta_A = \{\theta^g_A, \theta^h_A\}$. For subnetwork B, we can similarly obtain $f_B = g_B \circ h_B$ with $\theta_B = \{\theta^g_B, \theta^h_B\}$. Our goal is to accurately predict the semantic label of each pixel of test medical images using the training dataset $D$.

The overall framework of our proposed CSSML is depicted in Fig. 1. CSSML is composed of two subnetworks with entirely different architectures. We introduce a CDFL regularization to reduce the coupling of features extracted by the subnetworks and promote more diversified predictions. For the pseudo-labels generated by the two subnetworks, we employ the classic CPS regularization to perform cross-supervision on the non-conflicting pseudo-labels. For conflicting pseudo-labels, we introduce a GMPS regularization to assess their reliability based on geometric structures, and selectively leverage the more reliable pseudo-labels from the two subnetworks to supervise the other one. Below, we introduce the main components of CSSML, *i.e.*, CDFL regularization and GMPS regularization.

### 3.2. Conflict-aware distinct feature learning

Employing subnetworks $f_A$ and $f_B$ with distinct architectural designs helps mitigate the challenge of homogenization. However, medical images contain a variety of lesions that exhibit uneven intensities, adjacent boundaries, and irregular shapes. Different subnetworks may still generate coupled incorrect predictions for regions prone to missegmentation. To reduce the coupling between features extracted by the subnetworks and promote more diversified predictions, we introduce the CDFL regularization.

Specifically, a mini-batch of input data $M$ contains $n$ data $\mathbf{x}^M$, consisting of randomly sampled labeled images $\mathbf{x}^l$ from $D^l$ and randomly sampled unlabeled images $\mathbf{x}^u$ from $D^u$. The batch of images $\mathbf{x}^M$ is then fed into feature extractors $h_A(\cdot)$ and $h_B(\cdot)$, respectively. The generated features can be denoted as:

$$\mathbf{z}^l_A, \mathbf{z}^u_A = h_A(\mathbf{x}^M, \theta^h_A), \tag{1}$$

$$\mathbf{z}^l_B, \mathbf{z}^u_B = h_B(\mathbf{x}^M, \theta^h_B). \tag{2}$$

To implement feature discrepancy maximization, we introduce a simple projection function to map the features extracted by $h_B$ to another feature space:

$$\hat{\mathbf{z}}^l_B = \phi(\mathbf{z}^l_B), \hat{\mathbf{z}}^u_B = \phi(\mathbf{z}^u_B). \tag{3}$$

where the projection function $\phi(\cdot)$ consists of a simple $1 \times 1$ convolutional layer followed by a non-linear activation layer.

To reduce the coupling of features, the devised CDFL regularization seeks to minimize the cosine similarity between the features generated by the two subnetworks. The objective can be formally expressed as follows:

$$L^l_{CDFL} = 1 + \frac{\mathbf{z}^l_A \cdot \hat{\mathbf{z}}^l_B}{\left\|\mathbf{z}^l_A\right\| \times \left\|\hat{\mathbf{z}}^l_B\right\|}, L^u_{CDFL} = 1 + \frac{\mathbf{z}^u_A \cdot \hat{\mathbf{z}}^u_B}{\left\|\mathbf{z}^u_A\right\| \times \left\|\hat{\mathbf{z}}^u_B\right\|}, \tag{4}$$

where coefficient 1 ensures that the values of $L^l_{CDFL}$ and $L^u_{CDFL}$ always remain non-negative, facilitating the effective optimization of model parameters. Note that the CDFL regularization was applied only to subnetwork B in our setting.

### 3.3. Geometry-aware mutual pseudo supervision

Through the integration of two subnetworks with distinct architectures and CDFL regularization, we can obtain diversified predictions for regions prone to missegmentation. If the two subnetworks yield different outputs for the same pixel, it indicates that at least one of them must be incorrect. Cross pseudo supervision (CPS) utilizes pseudo-labels generated by one subnetwork to supervise the other subnetwork. This strategy utilizes potentially inaccurate pseudo-labels for cross supervision, which poses a risk of the two subnetworks overfitting to prediction errors and consequently stepping into collapse. To this end, we first assess the reliability of the predictions generated by the two subnetworks to ensure that only correct pseudo-labels are utilized for cross supervision.

It is evident that the consistent predictions derived from heterogeneous subnetworks with CDFL regularization suggest a high probability of correctness for the corresponding pseudo-labels. Therefore, we simply select the pseudo-label with the higher probability from the two subnetworks to supervise its counterpart. For conflicting pseudo-labels, we introduce GMPS regularization to evaluate their trustworthiness based on their geometric structures, as demonstrated in Fig. 2. Without loss of generality, for the conflicting predictions $\mathbf{p}^{u_2}_A$ and $\mathbf{p}^{u_2}_B$ of unlabeled image $\mathbf{x}^{u_2}$ (abbreviated as $u_2$), the feature representation $\mathbf{f}^{u_2}_B$ generated by subnetwork $f_B$ is farther from its class prototype $\mathbf{v}^{y'}_B$ than the corresponding feature representation $\mathbf{f}^{u_2}_A$ generated by subnetwork $f_A$. Consequently, the GMPS regularization takes the more reliable pseudo-label $\mathbf{y}^{u_2}_A$, derived from subnetwork $f_A$, to supervise subnetwork $f_B$. Our GMPS fundamentally differs from the confidence-based pseudo-label

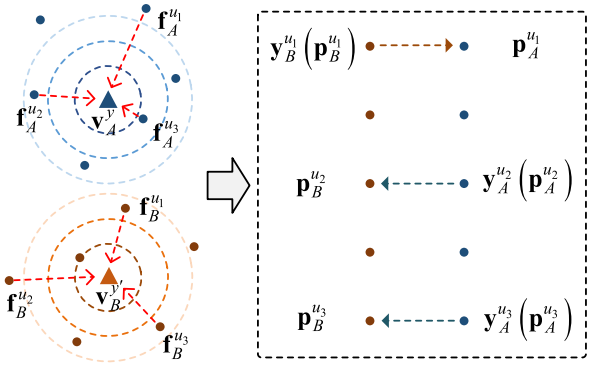


**Fig. 2.** Illustration of the proposed GMPS regularization.

selection strategy. The latter neglects the presence of conflicting predictions, which implies that a pseudo-label with higher confidence may still be incorrect. Above all, the proposed GMPS regularization selectively leverages the more reliable pseudo-labels from the two subnetworks to supervise its counterpart, thereby enhancing the overall robustness and reliability of the learning process.

Specifically, the probability outputs of subnetworks A and B can be obtained by using the pixel-wise softmax function $\sigma(\cdot)$, which can be represented as:

$$\mathbf{p}_A^l = \sigma\left(g_A\left(\mathbf{z}_A^l, \theta_A^g\right)\right), \mathbf{p}_A^u = \sigma\left(g_A\left(\mathbf{z}_A^u, \theta_A^g\right)\right), \tag{5}$$

$$\mathbf{p}_B^l = \sigma\left(g_B\left(\hat{\mathbf{z}}_B^l, \theta_B^g\right)\right), \mathbf{p}_B^u = \sigma\left(g_B\left(\hat{\mathbf{z}}_B^u, \theta_B^g\right)\right). \tag{6}$$

The predicted one-hot label of pixels from unlabeled images obtained from the two subnetworks can be expressed as follows:

$$\mathbf{y}_A^u = One\text{-}hot\left(\mathbf{p}_A^u\right), \mathbf{y}_B^u = One\text{-}hot\left(\mathbf{p}_B^u\right), \tag{7}$$

where $One\text{-}hot(\cdot)$ transforms the probability predictions $\mathbf{p}_A^u$ and $\mathbf{p}_B^u$ to hard labels $\mathbf{y}_A^u$ and $\mathbf{y}_B^u$. Therefore, the mask $\mathbf{M}$ of regions with conflicting pseudo-labels from an unlabeled image can be represented as follows:

$$\mathbf{M} = \mathbb{I}\left(\mathbf{y}_A^u \neq \mathbf{y}_B^u\right). \tag{8}$$

Here, $\mathbb{I}(\cdot)$ is the indicator function. Thus, the mask of regions with consistent pseudo-labels can be denoted as $(\mathbf{I} - \mathbf{M})$, in which $\mathbf{I}$ is the matrix with all elements equal to 1. We then select the pseudo-label with the higher probability value, which can be formulated as:

$$\mathbf{W} = \mathbb{I}\left(\max\left(\mathbf{p}_A^u\right) \leq \max\left(\mathbf{p}_B^u\right)\right). \tag{9}$$

Thus, for the regions with consistent pseudo-labels, we simply perform consistent cross-supervision (CCS) regularization for the two subnetworks as follows:

$$L_{Con}^A = (\mathbf{I} - \mathbf{M}) \odot \mathbf{W} \odot Dice\left(\mathbf{p}_A^u, \mathbf{y}_B^u\right), \tag{10}$$

$$L_{Con}^B = (\mathbf{I} - \mathbf{M}) \odot (\mathbf{I} - \mathbf{W}) \odot Dice\left(\mathbf{p}_B^u, \mathbf{y}_A^u\right). \tag{11}$$

Here, $Dice(\cdot)$ denotes the Dice loss. $\odot$ denotes the element-wise product operation.

If the pseudo-labels conflict, it means that one of them is incorrect. In such scenarios, selectively employing the more reliable pseudo-labels to provide supervision signal is conducive to averting performance degradation. As pixel complexity increases, the likelihood of generating unreliable pseudo-labels also rises. Hence, we utilize the intra-class semantic consistency of pixels based on their geometric structures to evaluate the reliability of pseudo-labels. As shown in Fig. 2, we calculate the class prototypes of each pixel from unlabeled images in the feature space obtained by subnetwork A and subnetwork B, which can be expressed as:

$$\mathbf{v}_A^y = \frac{1}{\left|\mathbb{I}\left(\mathbf{y}_A^u = y\right)\right|} \sum_{h,w} \mathbb{I}\left(\mathbf{y}_A^u = y\right) \odot \mathbf{f}_A^u, y = 1, 2, \cdots, Y, \tag{12}$$

$$\mathbf{v}_B^y = \frac{1}{\left|\mathbb{I}\left(\mathbf{y}_B^u = y\right)\right|} \sum_{h,w} \mathbb{I}\left(\mathbf{y}_B^u = y\right) \odot \mathbf{f}_B^u, y = 1, 2, \cdots, Y. \tag{13}$$

where $\mathbf{f}_A^u$ and $\mathbf{f}_B^u$ denote the feature maps from the penultimate layer of the two subnetworks. Then, the consistency of pixels and the $y$-th prototype can be measured as:

$$\mathbf{c}_A^y = \frac{\mathbf{f}_A^u \cdot \mathbf{v}_A^y}{\left\|\mathbf{f}_A^u\right\| \times \left\|\mathbf{v}_A^y\right\|}, \mathbf{c}_B^y = \frac{\mathbf{f}_B^u \cdot \mathbf{v}_B^y}{\left\|\mathbf{f}_B^u\right\| \times \left\|\mathbf{v}_B^y\right\|}. \tag{14}$$

Therefore, the mask $\mathbf{G}$ to reflect the more reliable pseudo-labels from the two subnetworks can be formulated as:

$$\mathbf{G}_A = \mathbb{I}\left(\mathbf{c}_A \leq \mathbf{c}_B\right), \tag{15}$$

$$\mathbf{c}_A = \max_y\left(\mathbf{c}_A^y\right), \mathbf{c}_B = \max_{y'}\left(\mathbf{c}_B^{y'}\right), y \neq y'. \tag{16}$$

With the reliability of pseudo-labels established, we can perform mutual pseudo supervision for regions with inconsistent predictions. Thus, the GMPS regularization for the two subnetworks can be formulated as:

$$L_{Incon}^A = \mathbf{M} \odot \mathbf{G}_A \odot Dice\left(\mathbf{p}_A^u, \mathbf{y}_B^u\right), \tag{17}$$

$$L_{Incon}^B = \mathbf{M} \odot \left(\mathbf{I} - \mathbf{G}_A\right) \odot Dice\left(\mathbf{p}_B^u, \mathbf{y}_A^u\right). \tag{18}$$

It is evident that the introduced GMPS regularization evaluates the reliability of pixel-wise predictions produced by the two subnetworks. This guarantees that only the reliable pseudo-labels are utilized to provide supervision signals, thereby preventing the model from overfitting to unreliable pseudo-labels in the training procedure.

**Algorithm 1:** Learning algorithm for CSSML.

**Input:** Labeled image datasets $D^l$, unlabeled image dataset $D^u$, trade-off coefficients $\beta$, $\lambda_C$ and $\lambda_U$, learning rate $\eta$, max iterations $T$.

**Output:** Model parameters $\theta_A$ and $\theta_B$ of subnetwork A and subnetwork B.

```
Initialize: Model parameters θ_A(0) and θ_B(0)
for t = 0, 1, ⋯, T − 1 do
    for each mini-batch M do
        Compute z_A^l, z_A^u, z_B^l and z_B^u using Eqs. (1)-(2),
        Compute ẑ_B^l and ẑ_B^u using Eq. (3),
        Compute L_CDFL^l and L_CDFL^u using Eq. (4),
        Compute p_A^l, p_A^u, p_B^l and p_B^u using Eqs. (5)-(6),
        Compute y_A^u and y_B^u using Eq. (7),
        Compute masks M and W using Eqs. (8)-(9),
        Compute L_Con^A and L_Con^B using Eqs. (10)-(11),
        Compute class prototypes v_A^y and v_B^y, ∀y,
        Compute c_A^y and c_B^y, ∀y,
        Compute masks G_A using Eqs. (15)-(16),
        Compute L_Incon^A and L_Incon^B using Eqs. (17)-(18),
        // Optimize model parameter for subnetwork A
        Compute supervised loss L_sup^A using Eq. (20),
        Compute unsupervised loss L_unsup^A using Eq. (22),
        Optimize θ_A(t) using Eq. (19),
        // Optimize model parameter for subnetwork B
        Compute supervised loss L_sup^B using Eq. (21),
        Compute unsupervised loss L_unsup^B using Eq. (23),
        Optimize θ_B(t) using Eq. (19).
    end
end
```

### 3.4. Objective function

Overall, the objective functions of subnetwork A and subnetwork B in the CSSML framework can be formulated as:

$$L_A = L_{sup}^A + L_{unsup}^A, L_B = L_{sup}^B + L_{unsup}^B, \tag{19}$$

**Table 1**
Performance comparison with different semi-supervised learning methods on ACDC dataset.

| Methods | Percentage | | Metrics | | | | |
|---|---|---|---|---|---|---|---|
| | Labeled | Unlabeled | DSC (%) | Jaccard (%) | Precision (%) | TPR (%) | 95HD (mm) |
| DeepLabv3+ Chen et al. (2018) | 100%(1312) | 0 | 90.09 | 82.31 | 89.61 | 91.05 | 1.06 |
| UNet Ronneberger et al. (2015) | 100%(1312) | 0 | 89.51 | 81.56 | 91.15 | 88.53 | 1.22 |
| DeepLabv3+ Chen et al. (2018) | 20%(262) | 0 | 82.28 | 70.92 | 82.89 | 83.40 | 5.32 |
| UNet Ronneberger et al. (2015) | 20%(262) | 0 | 76.87 | 64.71 | 82.74 | 76.33 | 5.00 |
| MT Tarvainen and Valpola (2017) | 20%(262) | 80%(1050) | 84.00(1.72) | 73.46(2.54) | 82.91(0.02) | 86.72(3.32) | 3.99(1.33) |
| EM Vu et al. (2019) | 20%(262) | 80%(1050) | 84.33(2.05) | 73.88(2.96) | 83.51(0.62) | 86.64(3.24) | 3.58(1.74) |
| DCT Qiao et al. (2018) | 20%(262) | 80%(1050) | 84.40(2.12) | 73.83(2.91) | 86.01(3.12) | 83.83(0.43) | 5.19(0.13) |
| DAN Zhang et al. (2017) | 20%(262) | 80%(1050) | 84.74(2.46) | 74.50(3.58) | 83.90(1.01) | 86.94(3.54) | 3.98(1.34) |
| UA-MT Yu et al. (2019) | 20%(262) | 80%(1050) | 84.78(2.50) | 74.52(3.60) | 84.58(1.69) | 86.36(2.96) | 3.79(1.53) |
| ICT Verma et al. (2022) | 20%(262) | 80%(1050) | 84.78(2.50) | 74.54(3.62) | 83.92(1.03) | 87.02(3.62) | 3.71(1.61) |
| CPS Chen et al. (2021b) | 20%(262) | 80%(1050) | 86.00(3.72) | 76.09(5.17) | 89.10(6.21) | 84.04(0.64) | 1.83(3.49) |
| MCF Wang et al. (2023a) | 20%(262) | 80%(1050) | 87.54(5.26) | 78.49(7.57) | 88.32(5.43) | 87.84(4.44) | 2.46(2.86) |
| PLSML Hang et al. (2025) | 20%(262) | 80%(1050) | 87.04(4.76) | 77.86(6.94) | 88.78(5.89) | 86.41(3.01) | 2.63(2.69) |
| 0.851, 0.851, 0.851 CSSML (Ours) | 20%(262) | 80%(1050) | **88.28(6.00)** | **79.64(8.72)** | **89.68(6.79)** | **87.87(4.47)** | **1.50(3.82)** |

where $L_{sup}^{A}$ and $L_{sup}^{B}$ denote the supervised losses for two networks, respectively. The supervised losses for two subnetworks are formulated as:

$$L_{sup}^{A} = Dice\left(\mathbf{p}_{A}^{l}, \mathbf{y}\right), \tag{20}$$

$$L_{sup}^{B} = Dice\left(\mathbf{p}_{B}^{l}, \mathbf{y}\right) + \beta \cdot L_{CDFL}^{l}. \tag{21}$$

Here, the trade-off parameter $\beta$ controls the strength of CDFL regularization. The unsupervised losses for the two subnetworks can be formulated as:

$$L_{unsup}^{A} = \lambda_C \cdot L_{Con}^{A} + \lambda_U \cdot L_{Incon}^{A}, \tag{22}$$

$$L_{unsup}^{B} = \lambda_C \cdot L_{Con}^{B} + \lambda_U \cdot L_{Incon}^{B} + \beta \cdot L_{CDFL}^{u}, \tag{23}$$

where parameters $\lambda_C$ and $\lambda_U$ are used to balance the consistent and inconsistent cross-supervision regularizations.

Finally, we summarize the thorough training process of the proposed CSSML framework in Algorithm 1.

## 4. Experiments

### 4.1. Datasets and pre-processing

*Automated Cardiac Diagnosis Challenge (ACDC)*[1]: This dataset Bernard et al. (2018) consists of 100 magnetic resonance imaging (MRI) scans collected from different patients. Each image was captured while the subject was holding their breath and encompasses the heart from its base to the apex of the left ventricle through a sequence of short-axis slices, each with a thickness ranging from 5 to 8 mm. The spatial resolution in the short-axis in-plane ranged from 0.83 to 1.75 $mm^2$ per pixel. Each scan was annotated and verified by specialists, containing three organs, *i.e.*, the right ventricle cavity, left ventricle cavity, and the myocardium. Following Huang et al. (2024), we utilized 70 scans for training, 10 scans for validation, and 20 scans for testing in our experiment. Additionally, we cropped the slices to a size of 256x256 for input.

*Synapse Multi-organ Segmentation Dataset (Synapse)*[2]: This dataset Landman et al. (2015) contains 30 computed tomography (CT) scans and involves eight abdominal organs (aorta, gallbladder, spleen, left kidney, right kidney, liver, pancreas, and stomach). Each CT volume consists of 85 ~ 198 slices, with each slice having a resolution of 512 × 512 pixels and a voxel spatial resolution of ([0.54 ~ 0.54] × [0.98 ~ 0.98] × [2.5 ~ 5.0]) $mm^3$. Following Huang et al. (2024), 18 scans and 12 scans were utilized for training and testing, respectively. In this dataset, we cropped the slices to a size of 320x320 for input.

*Polyp Segmentation Dataset (Polyp)*[3]: The publicly available polyp dataset CVC-ClinicDB Bernal et al. (2015) was used, which contains 612 colonoscopy images. Each image was paired with a binary ground truth label created by experts through the manual delineation of a mask over the region occupied by the polyp. For the experiment, 70% (429 images), 10% (61 images), and 20% (122 images) were allocated for training, validation, and testing, respectively. Additionally, the images were cropped to a size of 384x288 pixels for input. Table 2

### 4.2. Experimental setup

To validate the effectiveness of the proposed CSSML for semi-supervised medical image segmentation, we conducted a comparative analysis against several state-of-the-art SSL methods. The comparison methods are as follows: 1) MT Tarvainen and Valpola (2017); 2) ICT Verma et al. (2022); 3) DCT Qiao et al. (2018); 4) DAN Zhang et al. (2017); 5) UA-MT Yu et al. (2019); 6) EM Vu et al. (2019); 7) CPS Chen et al. (2021b); 8) MCF Wang et al. (2023a); 9) PLSML Hang et al. (2025); and 10) our proposed CSSML.

In the experiment, we re-implemented all comparison methods using their open-source codes and adopted DeepLabv3+ Chen et al. (2018) as the network backbone for the comparison methods MT, ICT, DCT, DAN, UA-MT, EM, and CPS. For the comparison methods MCF, PLSML, and our CSSML, we used UNet Ronneberger et al. (2015) and DeepLabv3+ as subnetwork A and subnetwork B, respectively. In our experiment, the subnetworks were trained using the SGD optimizer for 30K iterations, with the learning rate $\eta$ set to 0.01 for the ACDC dataset and 0.0125 for the Synapse dataset. Additionally, the number of iterations and the learning rate $\eta$ were set to 10K and 0.0001 for the Polyp dataset. For all three datasets, the batch size was set to 8, consisting of 4 labeled images and 4 unlabeled images. For CSSML, the trade-off coefficient $\beta$ was set to 0.1, 0.1, and 0.01 for the three medical image datasets, respectively. The trade-off coefficient $\lambda_C$ was set to 0.01, 0.005, and 0.01, respectively. The trade-off coefficient $\lambda_U$ was set to 0.1, 0.01, and 0.003, respectively. Besides, all other settings for the comparison methods remained consistent with those used in the original work. The implementation of the proposed CSSML framework was performed using PyTorch on one RTX 4090 GPU.

To comprehensively evaluate the segmentation performance of all comparison methods, we adopted the Dice similarity coefficient (DSC), Jaccard, Precision, 95% Hausdorff Distance (95HD), and True Positive Rate (TPR) as the evaluation metrics for all datasets in this paper.

[1] https://www.creatis.insa-lyon.fr/Challenge/acdc/databases.html
[2] https://www.synapse.org/Synapse:syn3193805/wiki/217789
[3] https://datasetninja.com/cvc-612

**Table 2**
Performance comparison with different semi-supervised learning methods on Synapse dataset.

| Methods | Percentage | | Metrics | | | | |
|---|---|---|---|---|---|---|---|
| | Labeled | Unlabeled | DSC (%) | Jaccard (%) | Precision (%) | TPR (%) | 95HD (mm) |
| DeepLabv3+ Chen et al. (2018) | 100%(2211) | 0 | 81.70 | 72.30 | 84.05 | 81.32 | 11.32 |
| UNet Ronneberger et al. (2015) | 100%(2211) | 0 | 80.01 | 70.74 | 83.01 | 81.37 | 17.24 |
| DeepLabv3+ Chen et al. (2018) | 20%(442) | 0 | 63.80 | 52.80 | 76.89 | 59.24 | 30.16 |
| UNet Ronneberger et al. (2015) | 20%(442) | 0 | 63.15 | 51.67 | 74.12 | 58.93 | 31.35 |
| MT Tarvainen and Valpola (2017) | 20%(442) | 80%(1769) | 66.43(2.63) | 55.46(2.66) | 81.29(4.40) | 60.68(1.44) | 17.31(12.85) |
| EM Vu et al. (2019) | 20%(442) | 80%(1769) | 67.32(3.52) | 56.79(3.99) | 81.18(4.29) | 62.49(3.25) | 20.53(9.63) |
| DCT Qiao et al. (2018) | 20%(442) | 80%(1769) | 68.50(4.70) | 58.29(5.49) | 81.01(4.12) | 64.77(5.53) | 21.17(8.99) |
| DAN Zhang et al. (2017) | 20%(442) | 80%(1769) | 69.25(5.45) | 58.62(5.82) | 80.61(3.72) | 64.01(4.77) | 18.58(11.58) |
| UA-MT Yu et al. (2019) | 20%(442) | 80%(1769) | 69.80(6.00) | 58.30(5.50) | 79.52(2.63) | 66.29(7.05) | 16.30(13.86) |
| ICT Verma et al. (2022) | 20%(442) | 80%(1769) | 70.22(6.42) | 59.20(6.40) | 78.33(1.44) | 67.98(8.74) | 22.61(7.55) |
| CPS Chen et al. (2021b) | 20%(442) | 80%(1769) | 70.31(6.51) | 60.19(7.39) | 82.84(5.95) | 66.27(7.03) | 17.48(12.68) |
| MCF Wang et al. (2023a) | 20%(442) | 80%(1769) | 70.72(6.92) | 61.06(8.26) | 81.68(4.79) | 68.30(9.06) | 16.31(13.85) |
| PLSML Hang et al. (2025) | 20%(442) | 80%(1769) | 71.43(7.63) | 61.69(8.89) | 82.71(5.82) | 68.01(8.77) | 16.55(13.61) |
| 0.851, 0.851, 0.851 CSSML (Ours) | 20%(442) | 80%(1769) | **72.40(8.60)** | **62.61(9.81)** | **84.03(7.14)** | **68.46(9.22)** | **15.88(14.28)** |

**Table 3**
Performance comparison with different semi-supervised learning methods on Polyp dataset.

| Methods | Percentage | | Metrics | | | | |
|---|---|---|---|---|---|---|---|
| | Labeled | Unlabeled | DSC (%) | Jaccard (%) | Precision (%) | TPR (%) | 95HD (mm) |
| DeepLabv3+ Chen et al. (2018) | 100%(429) | 0 | 83.91 | 76.95 | 90.22 | 81.66 | 19.58 |
| UNet Ronneberger et al. (2015) | 100%(429) | 0 | 78.80 | 70.87 | 85.59 | 78.40 | 34.58 |
| DeepLabv3+ Chen et al. (2018) | 20%(86) | 0 | 69.48 | 59.27 | 79.63 | 68.09 | 49.38 |
| UNet Ronneberger et al. (2015) | 20%(86) | 0 | 63.08 | 51.22 | 67.46 | 69.72 | 83.03 |
| MT Tarvainen and Valpola (2017) | 20%(86) | 80%(343) | 72.08(2.60) | 62.73(3.46) | 83.02(3.39) | 69.15(1.06) | 34.73(14.65) |
| EM Vu et al. (2019) | 20%(86) | 80%(343) | 74.27(4.79) | 64.50(5.23) | 87.87(8.24) | 70.28(2.19) | 41.15(8.23) |
| DCT Qiao et al. (2018) | 20%(86) | 80%(343) | 77.15(7.67) | 68.49(9.22) | 84.45(4.82) | 75.91(7.82) | 36.04(13.34) |
| DAN Zhang et al. (2017) | 20%(86) | 80%(343) | 75.72(6.24) | 66.01(6.74) | 87.02(7.39) | 71.97(3.88) | 32.84(16.54) |
| UA-MT Yu et al. (2019) | 20%(86) | 80%(343) | 77.99(8.51) | 68.88(9.61) | 88.92(9.29) | 75.73(7.46) | 36.60(12.87) |
| ICT Verma et al. (2022) | 20%(86) | 80%(343) | 78.68(9.20) | 70.02(10.75) | 89.36(9.73) | 76.75(8.66) | 33.79(15.41) |
| CPS Chen et al. (2021b) | 20%(86) | 80%(343) | 78.90(9.42) | 70.92(11.56) | 86.07(6.44) | 78.26(10.17) | 29.97(19.41) |
| MCF Wang et al. (2023a) | 20%(86) | 80%(343) | 80.26(10.78) | 72.33(13.06) | 85.85(6.22) | 78.77(10.68) | 26.31(23.07) |
| PLSML Hang et al. (2025) | 20%(86) | 80%(343) | 80.19(10.71) | 72.60(13.33) | 85.73(6.10) | 78.40(10.31) | 26.39(22.99) |
| 0.851, 0.851, 0.851 CSSML (Ours) | 20%(86) | 80%(343) | **81.26(11.78)** | **74.00(14.73)** | **89.77(10.14)** | **79.17(11.08)** | **26.05(23.33)** |

### *4.3. Comparison with state-of-the-Art methods*

Tables 1–3 list the segmentation performance of all SSL comparison methods using 20% labeled images and 80% unlabeled images in the training set across three medical image datasets. For the ACDC, Synapse, and Polyp datasets, there are approximately 262, 442, and 86 labeled slices, along with 1050, 1769, and 343 unlabeled slices, respectively. Moreover, the segmentation results derived from training fully supervised models, *i.e.*, UNet and DeepLabv3+, with 100% labeled data are regarded as the upper-bound performance. Meanwhile, the segmentation results obtained by training UNet and DeepLabv3+ with only 20% labeled data are regarded as the baseline performance. The best segmentation results for each evaluation metric are emphasized in bold. We also list the performance enhancements achieved by all SSL methods relative to the better baseline method, DeepLabv3+, which are presented in parentheses. Based on the experimental results depicted in Tables 1–3, several insights can be derived.

Overall, SSL methods consistently outperform the baseline methods UNet and DeepLabv3+ by utilizing extensive unlabeled images. Furthermore, the comparison methods MCF and PLSML achieve better segmentation performance than other SSL approaches in most cases. This verifies the efficacy of integrating two subnetworks with distinct architectures to prevent them from becoming coupled together. The proposed CSSML framework demonstrates superior segmentation performance compared with other SSL methods across all evaluation metrics. This confirms the effectiveness of integrating different subnetworks with the proposed CDFL and GMPS regularizations in the SSL framework.

Generally, CSSML exhibits promising segmentation results across three medical image datasets. Compared to the baseline method DeepLabv3+, CSSML demonstrates an average improvement of 8.79%, 11.09%, 8.02%, 8.26%, and 13.81% on DSC, Jaccard, Precision, TPR and 95HD, respectively. Furthermore, CSSML surpasses the average segmentation performance of the most competitive method MCF by margins of 1.14%, 1.46%, 2.54%, 0.20%, and 0.55% on DSC, Jaccard, Precision, TPR and 95HD, respectively. Specifically, for the ACDC dataset, CSSML outperforms the most competitive method MCF by 0.74%, 1.15%, 1.36%, 0.03%, and 0.96% on DSC, Jaccard, Precision, TPR and 95HD, respectively. For Synapse dataset, the segmentation results of CSSML are 0.97%, 0.92%, 1.32%, 0.45% and 0.67% higher than those of most competitive method PLSML, respectively. Similarly, for the Ploy dataset, CSSML achieves improvements of 1.00%, 1.67%, 3.92%, 0.40%, and 0.26% over MCF, respectively. It is noteworthy that the segmentation performance of CSSML closely approaches the upper-bound performance of DeepLabv3+, in terms of the Precision metric for all datasets. Besides, CSSML achieves better segmentation performance than the upper-bound performance of UNet in many cases. These results verify that the proposed CSSML method can improve segmentation performance by selectively leveraging the more reliable pseudo-labels from the two subnetworks to supervise the other one.

Fig. 3 illustrates a visual comparison of segmentation performance between our CSSML and the comparison SSL methods on randomly selected slices from three medical image datasets. Fig. 3(a) demonstrates the segmentation results for the ACDC dataset. The first eight columns display the segmentation results derived from the comparison SSL methods, whereas the last two columns show the segmentation results of our CSSML method and the ground truth (GT). As shown, most of the comparison SSL methods fail to accurately segment the right ventricle (RV) cavity (blue annotations), while our CSSML achieves better overall

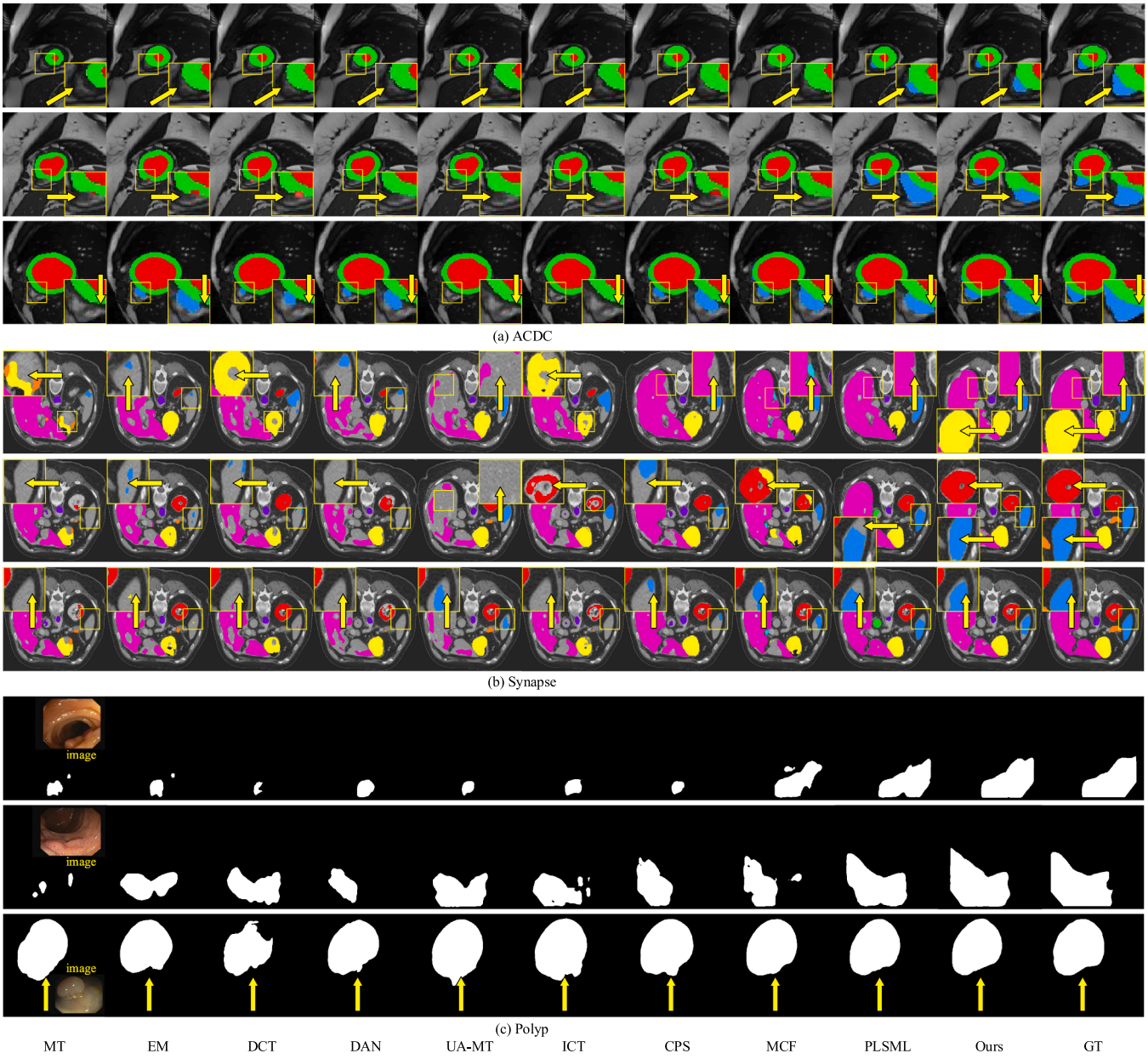



**Fig. 3.** Visualization of the segmentation results for qualitative comparison against other SSL methods: MT Tarvainen and Valpola (2017), EM Vu et al. (2019), DCT Qiao et al. (2018), DAN Zhang et al. (2017), UA-MT Yu et al. (2019), ICT Verma et al. (2022), CPS Chen et al. (2021b), MCF Wang et al. (2023a), PLSML Hang et al. (2025) in the three medical image datasets. We use the boxes to indicate the obviously worse segmentation results of the comparison methods. The zoom-in views of the boxed areas and the arrows highlight the regions worth noting.

segmentation performance. For the Synapse dataset, the presence of organs with heterogeneous intensities, adjacent boundaries, or irregular shapes makes it evident that the proposed CSSML performs better in segmenting some organs when contrasted with other SSL methods, particularly in the segmentation of the Liver (indicated by pink annotations), Spleen (blue annotations), Stomach (yellow annotations), and Left Kidney (red annotations). Similarly, for the Ploy dataset, our CSSML obtains better segmentation performance than other comparison SSL methods, as shown in Fig. 3(c). This observation further validates the benefits of combining different subnetworks with CDFL regularization and GMPS regularization, thereby helping the model avoid cognitive bias.

**Table 4**
Computational efficiency comparison among MT, MCF and our CSSML in terms of FLOPs (G), model parameters (M), GPU memory (GB), throughput (images/s), and mean latency (ms).

| Methods | FLOPs | Parameters | Memory | Throughput | Latency |
|---|---|---|---|---|---|
| MT | 24.0 | 59.5 | 0.50 | 520 | 15.4 |
| MCF | 65.6 | 93.9 | 0.61 | 367 | 21.8 |
| CSSML | 65.6 | 93.9 | 0.61 | 401 | 19.9 |

We further evaluated the computational efficiency of the proposed CSSML in comparison with two representative categories of SSL methods, namely MT and MCF, in terms of FLOPs (G), model parameters (M), GPU memory (GB), throughput (images/s), and mean latency (ms). The corresponding results are summarized in Table 4. Since certain metrics are sensitive to input size, we report the results only on the ACDC dataset without loss of generality. Note that both the most competitive baseline, MCF, and our CSSML adopt UNet and DeepLabv3+ as subnet-

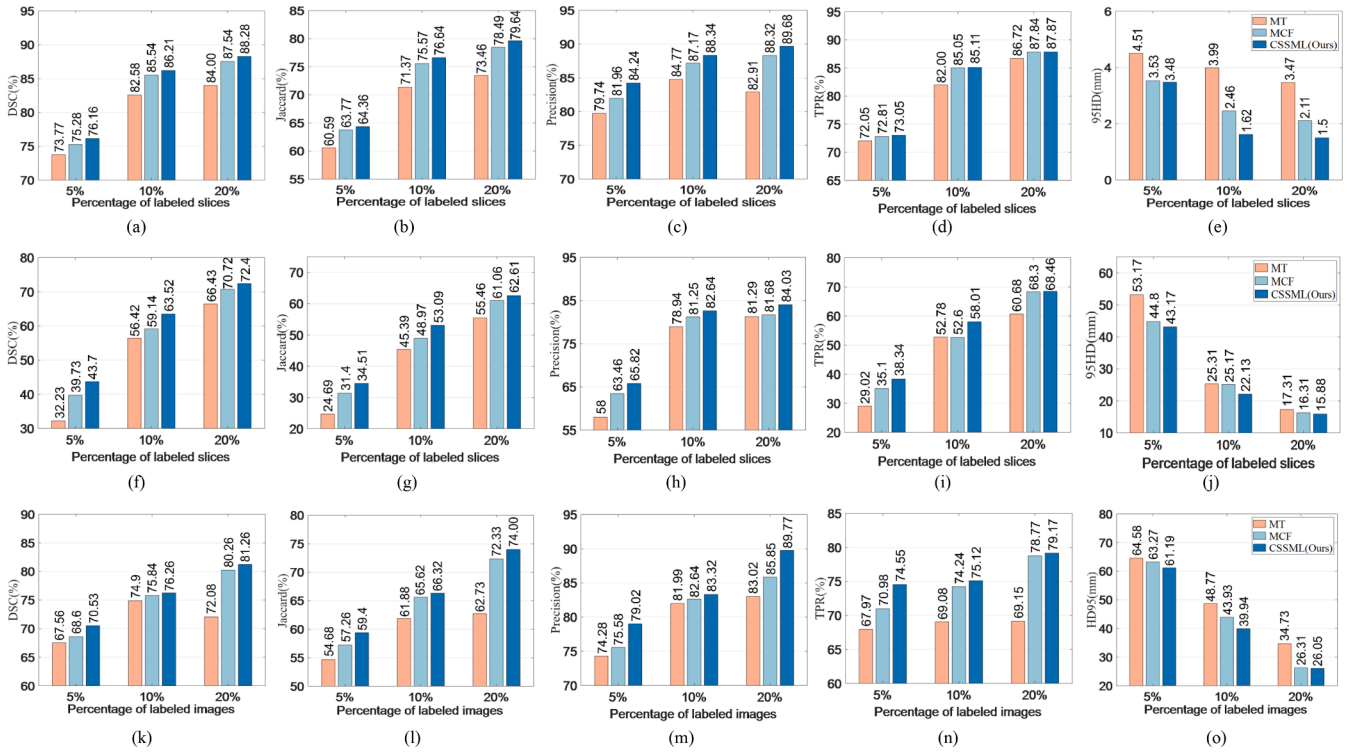



**Fig. 4.** Segmentation performance of baseline SSL method MT, most competitive method MCF and our CSSML when using different percentage of labeled slices on three medical image datasets, (a)-(e) denotes the segmentation performance on ACDC dataset. (f)-(j) denotes the segmentation performance on Synapse dataset. (k)-(o) denotes the segmentation performance on Ploy p dataset.

**Table 5**
Ablation study of different components on the segmentation performance of ACDC and Synapse datasets.

| UNet | DeepLabv3+ | CPS | CDFL | GMPS | ACDC | | | | | Synapse | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | DSC | Jaccard | Precision | TPR | 95HD | DSC | Jaccard | Precision | TPR | 95HD |
| ✓ | | | | | 76.87 | 64.71 | 82.74 | 76.33 | 5.00 | 63.15 | 51.67 | 74.12 | 58.93 | 31.35 |
| | ✓ | | | | 82.28 | 70.92 | 82.89 | 83.40 | 5.32 | 63.80 | 52.80 | 76.89 | 59.24 | 30.16 |
| ✓ | ✓ | ✓ | | | 86.87 | 77.61 | 88.97 | 86.03 | 2.94 | 69.21 | 58.93 | 80.78 | 66.23 | 25.44 |
| ✓ | ✓ | ✓ | ✓ | | 87.31 | 78.21 | 88.93 | 86.76 | 2.44 | 70.39 | 59.82 | 77.99 | 67.73 | 20.57 |
| ✓ | ✓ | | ✓ | ✓ | 88.28 | 79.64 | 89.68 | 87.87 | 1.50 | 72.40 | 62.61 | 84.03 | 68.46 | 15.88 |

**Table 6**
Sensitivity of trade-off coefficient $\beta$ on the segmentation performance of ACDC and Synapse datasets.

| $\lambda_C = 0.01$ | ACDC | | | | | $\lambda_C = 0.005$ | Synapse | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $\lambda_U = 0.1$ | DSC | Jaccard | Precision | TPR | 95HD | $\lambda_U = 0.01$ | DSC | Jaccard | Precision | TPR | 95HD |
| $\beta = 0.001$ | 87.47 | 78.27 | 88.8 | 87.01 | 2.72 | $\beta = 0.001$ | 70.24 | 60.30 | 83.49 | 65.93 | 17.95 |
| $\beta = 0.01$ | 87.55 | 78.49 | 88.94 | 87.14 | 2.13 | $\beta = 0.01$ | 70.77 | 61.22 | 84.00 | 66.80 | 16.68 |
| $\beta = 0.1$ | **88.28** | **79.64** | **89.68** | **87.87** | **1.50** | $\beta = 0.1$ | **72.40** | **62.61** | 84.03 | **68.46** | **15.88** |
| $\beta = 1$ | 87.52 | 78.47 | 88.54 | 87.57 | 1.97 | $\beta = 0.5$ | 69.31 | 59.70 | **84.77** | 67.56 | 17.52 |
| $\beta = 10$ | 86.33 | 76.98 | 87.92 | 85.58 | 2.09 | $\beta = 1$ | 68.69 | 59.42 | 83.85 | 66.31 | 19.81 |

**Table 7**
Sensitivity of trade-off coefficient $\lambda_C$ on the segmentation performance of ACDC and Synapse datasets.

| $\beta = 0.1$ | ACDC | | | | | $\beta = 0.1$ | Synapse | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $\lambda_U = 0.1$ | DSC | Jaccard | Precision | TPR | 95HD | $\lambda_U = 0.01$ | DSC | Jaccard | Precision | TPR | 95HD |
| $\lambda_C = 0.001$ | 87.46 | 78.45 | 88.9 | 86.91 | 2.35 | $\lambda_C = 0.001$ | 71.03 | 61.35 | 83.25 | 66.31 | 17.85 |
| $\lambda_C = 0.01$ | **88.28** | **79.64** | **89.68** | **87.87** | **1.50** | $\lambda_C = 0.003$ | 71.92 | 61.91 | 83.56 | 66.73 | 16.67 |
| $\lambda_C = 0.1$ | 86.83 | 77.48 | 87.25 | 87.58 | 1.74 | $\lambda_C = 0.005$ | **72.40** | **62.61** | **84.03** | **68.46** | **15.88** |
| $\lambda_C = 0.5$ | 86.44 | 76.87 | 86.38 | 87.35 | 1.92 | $\lambda_C = 0.007$ | 71.24 | 61.48 | 83.54 | 68.19 | 23.31 |
| $\lambda_C = 1$ | 86.24 | 76.53 | 86.23 | 86.75 | 2.83 | $\lambda_C = 0.009$ | 71.03 | 60.68 | 83.33 | 67.24 | 24.57 |

**Table 8**
Sensitivity of trade-off coefficient $\lambda_U$ on the segmentation performance of ACDC and Synapse datasets.

| $\beta = 0.1$ | ACDC | | | | | $\beta = 0.1$ | Synapse | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $\lambda_C = 0.01$ | DSC | Jaccard | Precision | TPR | 95HD | $\lambda_C = 0.005$ | DSC | Jaccard | Precision | TPR | 95HD |
| $\lambda_U = 0.001$ | 87.39 | 78.29 | 88.92 | 86.61 | 2.46 | $\lambda_U = 0.001$ | 71.64 | 61.99 | 83.80 | 67.41 | 18.37 |
| $\lambda_U = 0.01$ | 87.54 | 78.53 | 89.46 | 86.64 | 2.37 | $\lambda_U = 0.01$ | **72.40** | **62.61** | 84.03 | **68.46** | **15.88** |
| $\lambda_U = 0.1$ | **88.28** | **79.64** | **89.68** | **87.87** | **1.50** | $\lambda_U = 0.1$ | 71.89 | 62.03 | **84.76** | 67.93 | 16.78 |
| $\lambda_U = 1$ | 87.58 | 78.62 | 86.91 | 87.24 | 2.03 | $\lambda_U = 0.5$ | 71.29 | 61.43 | 84.32 | 66.81 | 18.70 |
| $\lambda_U = 10$ | 86.34 | 76.82 | 86.79 | 86.26 | 2.45 | $\lambda_U = 1$ | 70.89 | 61.72 | 84.25 | 66.27 | 20.30 |

works, whereas MT employs identical subnetworks (*i.e.*, DeepLabv3+). As shown in Table 4, SSL methods using identical subnetworks generally require less GPU memory, exhibit lower latency, and achieve higher inference throughput. Although CSSML introduces higher computational costs compared with these methods, it yields substantial performance gains with a reasonable overhead. Moreover, compared with the strongest baseline, MCF, CSSML achieves superior segmentation performance while maintaining acceptable and comparable computational overhead, demonstrating its overall effectiveness.

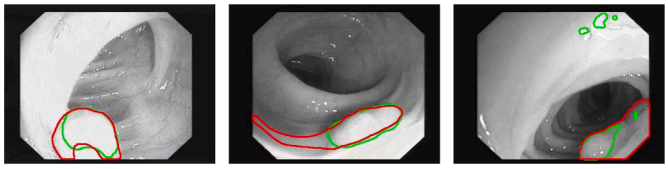


**Fig. 5.** Visualization of failure cases. Green lines denote our segmentation results, and red lines for ground truth.

### 4.4. Analysis of the proposed CSSML framework

#### 4.4.1. Influence of different percentage of labeled data

We investigated the impact of varying percentages of labeled images in our SSL method. We performed experiments using 5%, 10%, and 20% of labeled images, as shown in Fig. 4. To comprehensively evaluate the segmentation results obtained by our CSSML, we compared it with the baseline method MT and the most competitive SSL method MCF. It is observed that the segmentation performance of all three SSL methods improves as the amount of labeled data increases. The segmentation results approach the upper-bound performance of the fully supervised method in some metrics. Overall, across all metrics, the proposed CSSML framework consistently surpasses the performance of the other two comparison methods on the three medical image datasets. These segmentation results demonstrate the efficacy of our approach in leveraging unlabeled data to achieve significant performance improvements. Moreover, we found that our CSSML trained on 10% labeled data achieves segmentation performance comparable to that of the comparison methods MT and MCF trained with 20% labeled data. This verifies that the synergistic learning between CDFL and GMPS regularizations helps the model avoid cognitive bias.

#### 4.4.2. Efficacy of different components

To gain better insight into the performance of the proposed CSSML framework, we conducted an ablation study to investigate the effectiveness of different components in CSSML. The results of the ablation study are listed in Table 5. The ablation study includes the following models: a) backbone network UNet (Scenario 1); b) backbone network DeepLabv3+ (Scenario 2); c) subnetworks UNet and DeepLabv3+ with CPS regularization (Scenario 3); d) subnetworks UNet and DeepLabv3+ with CPS and CDFL regularizations (Scenario 4); e) subnetworks UNet and DeepLabv3+ with CDFL and GMPS regularizations (Scenario 5, CSSML). As demonstrated in Table 5, Scenario 3 yields better segmentation performance than Scenario 1 and Scenario 2, which demonstrates the effectiveness of using different subnetworks with CPS regularization. Moreover, Scenario 4 achieves better segmentation performance than Scenario 3, which verifies that CDFL can reduce feature coupling to further promote more diversified predictions, thereby preventing the two subnetworks from collapsing into each other. Compared to Scenario 4, Scenario 5 (CSSML) further enhances the segmentation performance. This observation highlights the importance of selectively utilizing the more reliable pseudo-labels from the two subnetworks to provide supervision to their counterpart, which contributes to enhancing the overall robustness and reliability of the model.

#### 4.4.3. Impact of hyper-parameters

We further investigate the influence of hyper-parameters on the segmentation performance of the proposed CSSML. Three hyper-parameters, $\beta$, $\lambda_C$, and $\lambda_U$, are involved in CSSML, in which $\beta$ controls the strength of CDFL loss, and $\lambda_C$ and $\lambda_U$ are used to balance the consistent and inconsistent cross-supervision regularizations, respectively. In the experiment, we kept two parameters constant while varying the third parameter to observe the segmentation performance of CSSML, as shown in Tables 6–8. Specifically, the parameter $\beta$ influences the generation of reasonably diversified predictions in the SSL model. When $\lambda_C = 0.01$ and $\lambda_U = 0.1$, the segmentation performance of CSSML on the ACDC dataset improves with the increase of $\beta$, which verifies CDFL regularization brought gains to the segmentation results. As shown in Table 6, when $\beta$ continues to increase, the segmentation performance degrades, indicating that neglecting other losses may adversely affect the segmentation performance. Similar results for parameters $\lambda_C$ and $\lambda_U$ are obtained in Tables 7–8. The impact of hyper-parameters on the segmentation performance of the Synapse dataset is similar to that of the ACDC dataset.

## 5. Discussion

We further analyzed several failure cases to better understand the limitations of the proposed model. As shown in Fig. 5, these cases mainly involve lesions with highly irregular shapes. In medical image segmentation tasks, most SSL models struggle to accurately delineate the boundaries of such irregular anatomical structures. This limitation largely stems from their reliance on large amounts of high-quality labeled medical images. For CSSML, irregular lesion shapes introduce additional challenges for the GMPS regularization when evaluating the reliability of pseudo-labels. Specifically, a single prototype may sometimes fail to adequately capture the feature diversity of all pixels within irregular regions, thereby affecting the accuracy of pseudo-label reliability estimation. To address this challenge, adopting a multi-prototype strategy could be a promising direction to improve both the accuracy and robustness of pseudo-label reliability evaluation.

For the cognitive bias issue, as discussed in Subsection 2.2, it refers to the phenomenon where a model persistently produces incorrect predictions due to overfitting to erroneous supervision signals. The proposed CSSML framework integrates CDFL and GMPS regularizations, whose synergistic learning helps the model effectively mitigate cognitive bias. Specifically, the CDFL regularization encourages two distinct subnetworks to learn diverse feature representations from the same input. With these diversified features, existing CPS- and consistency-based

regularizations expect the subnetworks to mutually learn from each other to enhance the model's generalization ability Wang et al. (2023b). Although effective, such approaches may still involve incorrect predictions as supervision signals, causing the model to overfit to unreliable pseudo-labels and consequently suffer from cognitive bias. Essentially, these models lack a self-correction mechanism to reduce the influence of inaccurate pseudo-labels. In light of this, we introduce the GMPS regularization to identify reliable pseudo-labels for supervision. For consistent predictions between the two subnetworks, we select the pseudo-label with higher confidence to supervise its counterpart. For conflicting predictions, reliable pseudo-labels are identified based on their geometric structures and are then used to supervise the other subnetwork. Overall, CDFL regularization promotes prediction diversity, while GMPS regularization identifies reliable pseudo-labels. Working synergistically, CDFL and GMPS equip the model with a find-and-correct capability, thereby effectively mitigating cognitive bias. This effectiveness is further validated by the ablation study results presented in Table 5.

In terms of computational cost, compared to the most competitive method, MCF, the proposed CSSML introduces only a lightweight projection function consisting of a simple $1 \times 1$ convolutional layer followed by a non-linear activation layer, along with a similarity computation between features and class prototypes that has linear complexity. Therefore, the additional computational cost introduced by CSSML is negligible compared to MCF. Moreover, as shown in Table 6–8, the optimal hyperparameters of CSSML on the ACDC and Synapse datasets are $\beta$= 0.1, $\lambda_C = 0.01$, $\lambda_U = 0.1$ and $\beta$= 0.1, $\lambda_C = 0.005$, $\lambda_U = 0.01$, respectively. The optimal parameter $\beta$ remains consistent across both datasets. The differences in parameters $\lambda_C$ and $\lambda_U$ are relatively small, indicating that the optimal hyperparameters of CSSML are stable across different datasets. Nevertheless, developing efficient dataset-specific hyperparameter tuning strategies remains an important direction for future work, as it would enable the model to be rapidly adapted and deployed in real-world clinical environments.

Recent studies have explored text-guided and vision-language paradigms as a promising direction for medical image segmentation Zeng et al. (2025b). The proposed CSSML focuses on vision-based dual-network mutual learning, it is conceptually compatible with text-guided supervision. Textual information could be incorporated as an auxiliary modality to further regularize feature representations or to assist in identifying reliable pseudo-labels. Exploring the integration of text-guided learning with conflict-aware mutual learning represents a promising direction for future work and may further reduce the annotation burden in practical medical imaging applications.

## 6. Conclusions

In this paper, we present a novel conflict-aware semi-supervised mutual learning framework, termed CSSML, aimed at facilitating semi-supervised medical image segmentation by exploiting conflict-aware subnetworks while addressing conflicting pseudo-labels. The proposed CSSML employs two subnetworks with distinct architectures and incorporates two innovative regularizations: the CDFL regularization and the GMPS regularization. The CSSML framework can diminish the coupling of features extracted by the subnetworks, thereby promoting more diversified predictions. Furthermore, it has the capability to selectively utilize the more reliable pseudo-labels from the two subnetworks, thus offering effective supervision to its counterpart. The synergistic learning between the two regularizations facilitates the subnetworks in selectively incorporating the reliable knowledge imparted by the other subnetwork, thereby helping the model avoid cognitive bias. Experimental results on three medical image datasets demonstrate that CSSML achieves average improvements of 1.14%, 1.46%, and 2.54% in DSC, Jaccard, and Precision, respectively, over the strongest baseline, MCF, while using 80% fewer labeled samples than fully supervised methods. These results highlight the effectiveness of CSSML in enhancing segmentation performance and reducing the annotation burden in clinical practice.

Despite the promising performance of CSSML, there remains room for further improvement. For instance, the GMPS regularization currently evaluates the reliability of pseudo-labels based on the similarity between pixels and their corresponding prototypes. Incorporating more advanced similarity measurement strategies could further enhance the segmentation performance of medical images. Moreover, since the proposed CSSML is developed under the assumption of independent and identically distributed (IID) data, extending it to non-IID scenarios (*e.g.*, training and testing datasets obtained from different populations, equipment, or environments) is an important research direction. We plan to address these challenges in our future work.

## Credit Author Statement

**Wenlong Hang:** Conceptualization, Methodology, Validation, Formal analysis, Writing – review & editing; **Beijing Wang:** Data processing, Investigation, Visualization; **Shuang Liang:** Resources, Formal analysis, Writing – review & editing; **Qing Feng Zhang:** Supervision, Validation, Funding acquisition; **Qiang Wu:** Supervision, Resources, Project administration, Funding acquisition; **YuKun Jin:** Data curation, Investigation, Visualization; **Qiong Wang:** Resources, Formal analysis, Writing – review & editing; **Jin Qin:** Methodology, Validation, Project administration.

## Data availability

Data will be made available on request.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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